Comments regarding Presentation

- We had to give a 5-minute presentation on any topic we chose for the final exam. If you are short on time, I'd recommend going deeper on whatever you covered in your poster presentation.
- Putting real effort into the presentation pays off, it genuinely helped me during the questioning round and raised the overall quality of my answers.
- The first 10 slides are what I actually presented at the exam. The remaining slides were backup for the Q&A. I skipped the limitations slide due to the strict 5-minute limit, so be aware they enforce it tightly.
- One general tip: make sure you fully understand every diagram they show in the course. At the end of the oral exam, the examiners will pick one to three of them and ask you to walk through them in detail.

Final Exam - Materials Technology

Can theoretical tools predict the mechanical properties of PC/CF composites for 3D printing and injection moulding?

Presented by Younis Al-Atrash

3D-Printed Filament from Prusa

Recycled carbon fibre, discontinuous

Prusament PC Blend Carbon Fiber

Prusament PC Blend Carbon Fiber (PCCF) is a PC Blend filled with carbon fibers to improve its strength, toughness and temperature resistance. Unlike the unmodified PC Blend, PCCF comes with great dimensional stability, good resistance to UV light and common chemicals, better tensile strength and toughness, but most importantly with high-temperature resistance. It's optimal for printing technical components requiring great strength and high temperature tolerance.

Injection-Moulded Composites from RTP Co.

Virgin carbon fibre, discontinuous, 10% mas, ≈7% vol

RTP Co.

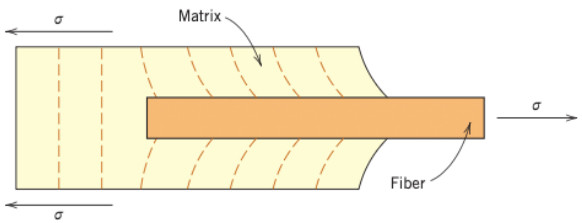
Imagineering Plastics®

The formulas I want to test

d is diameter of fiber, τ_c is shear stress of matrix, σ_m matrix stress, σ_f UTS of fiber, E elongation, V is vol%, K is fiber efficiency parameter from (0-1)

Critical fiber length

$$l_c = \frac{\sigma_f^* d}{2\tau_c}$$



Longitudinal Fracture Strength

L<l_c

$$\sigma_{cd'}^* = \frac{l\tau_c}{d} V_f + \sigma_m'(1 - V_f)$$

L>l_c

$$\sigma_{cd}^* = \sigma_f^* V_f \left(1 - \frac{l_c}{2l}\right) + \sigma_m'(1 - V_f)$$

discontinous & aligned fibers

Elongation of Composite

$$E_{cd} = KE_f V_f + E_m V_m$$

discontinous & randomly alligned fibers

Theoretical values

Measure	Theory Prusa	Actual Prusa	Theory RTP	Actual RTP
Fiber length (μm)	500 ideal*	300	645 ideal*	?
Fracture Strength[1,3] (MPa)	126	70 ± 3	197	107
Modulus of Elasticity[2] (GPa)	3.82-11.9	3.2 ± 0.1	3.82-11.9	7.58
K-Value	0.061	?	0.33	?

[1] Westimate for RTP that the fiber length is exactly the critical fiber length.

[2] We take the books range of K being from 0.1 - 0.6

[3] We assume 7% vol frac CF for Prusa’s material

Discussion – What can we learn

Prusa filament

- Nozzle size is $400\ \mu m$, $L < l_c$
- Avg. fiber length $\mathrm{L} = 300\ \mu m$

$$\sigma_f^* = \frac{L \cdot 2\tau_c}{d} \qquad \sigma_f^* = \frac{300\mu m \cdot 2 \cdot 22.8MPa}{7\mu m} = 1{,}954\ MPa$$

Effective strength: 1,954 MPa;

Potential Wasted: 1,346 MPa wasted.

RTP filament

According to Calister k values of $\sim$0.375 means fibers randomly and uniformly distributed within a specific plane.

This can be explained for RTP from two effects:

1) Skin-Core effect
2) Thin-Wall constraint from the dogbone

Limitation

Fracture Strength Formula

- Formula assumes aligned fibers, Prusa and RTP are likely randomly oriented

Elasticity Formula

- Voids are not accounted for in the elasticity formula
- Formula assumes $L \geq lc$ → not satisfied for Prusa's material

Material assumptions

- Prusament uses a PC blend + CF, not pure PC/CF → skews comparison
- Assumed 7% CF volume fraction for 3D print filament

Conclusion

It’s hard to know how well the formulas did in comparison to the real practice as some numbers had to be assumed and could largely fluctuate the values.

- Overall deviation of fracture strength ~45% for both production
- Good tool for spotting single analysis points
- Theoretical values can serve as a useful upper-bound estimate of efficiency

Questions

1. Why Polycarbonate?

2. Recycled vs. virgin CF, why does recycled have lower σ*_f (3,300 vs 4,200 MPa)?
    1. Should Prusa use virgin CF?

3. What were some of the limitations?

Sources

Callister, W. D., Jr., & Rethwisch, D. G. (2018). Materials science and engineering: An introduction (10th ed.). Wiley.

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Semitekolos, D., Terzopoulou, S., Zecchi, S., Marinis, D., Farsari, E., Amanatides, E., Sajdak, M., Sobek, S., Smok, W., Tański, T., Werle, S., Tagliaferro, A., & Charitidis, C. (2024). Performance restoration of chemically recycled carbon fibres through surface modification with sizing. Polymers, 17(1), 33. https://doi.org/10.3390/polym17010033

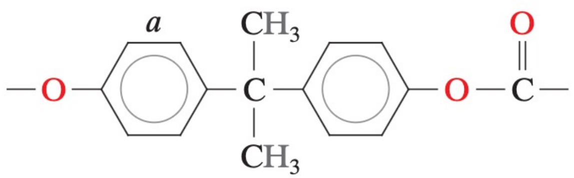
Cantrell, J., Rohde, S., Damiani, D., Gurnani, R., DiSandro, L., Anton, J., Young, A., Jerez, A., Steinbach, D., Kroese, C., & Ifju, P. (2016). Experimental characterization of the mechanical properties of 3D printed ABS and polycarbonate parts. In Conference Proceedings of the Society for Experimental Mechanics Series (pp. 89–105). Springer. https://doi.org/10.1007/978-3-319-41600-7_11

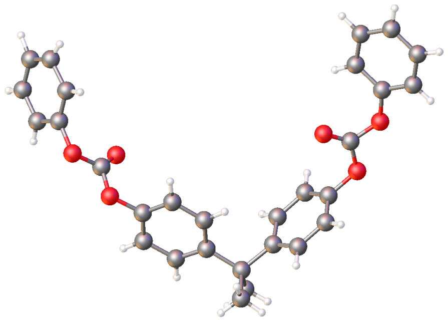
RTP Company. (n.d.). Series 300: Structural polycarbonate compounds technical data sheets. https://www.rtpcompany.com/technical-info/data-sheets/series-300/

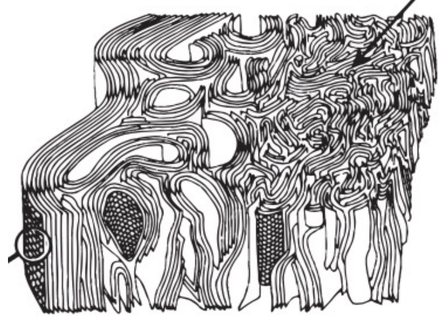
What is Polycarbonate (PC)? | What is Carbon Fibre (CF)?

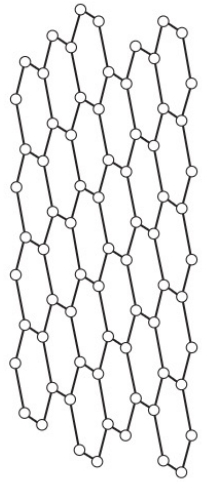
- Amorphous thermoplastic
- High impact resistance; low scratch resistance
- Made by condensation polymerisation
- Common additive: UV stabilisers

- Built from stacked graphene layers
- Graphitic (high stiffness) vs turbostratic (high strength)
- Small-diameter fibres with high strength and stiffness
- Primarily used as composite reinforcement







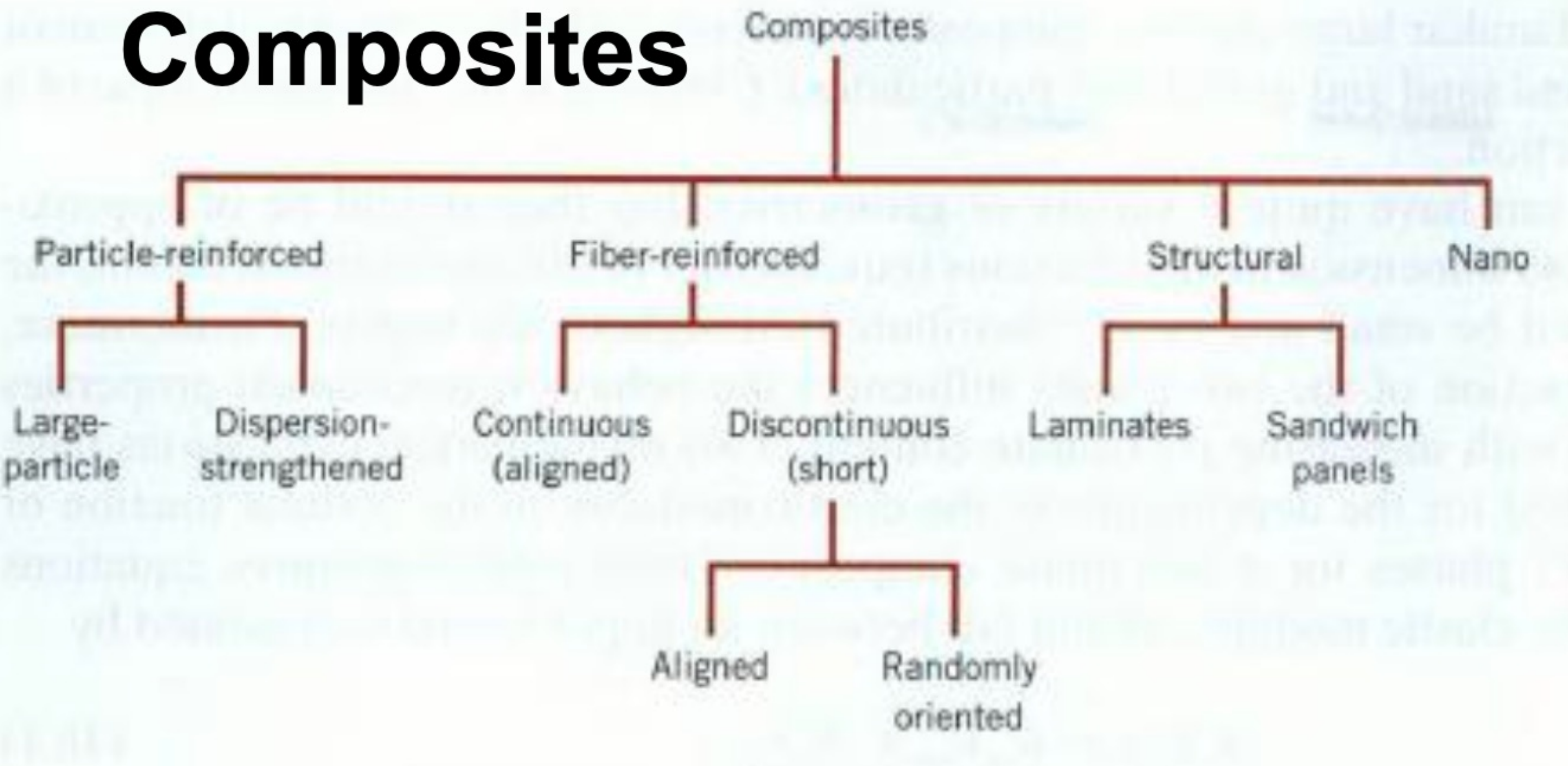


DTU

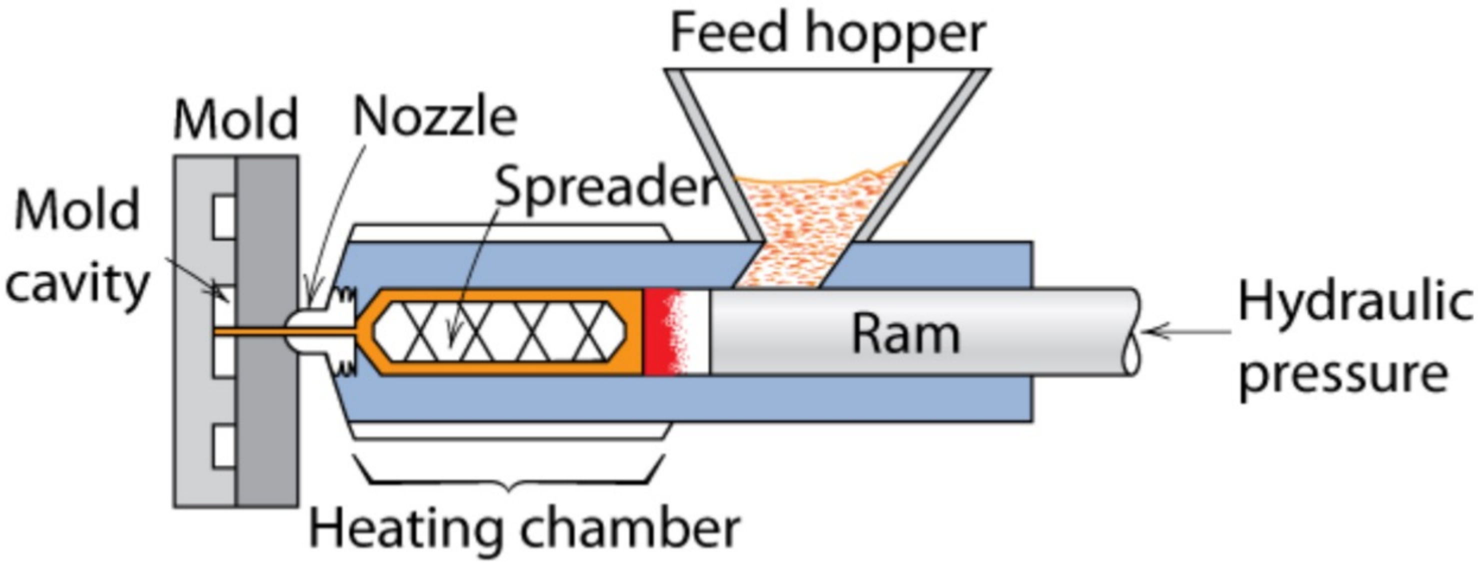
	Yield Strength (MPa)	Fracture toughness (MPa*m^(1/2))	Density (g/cm3)	Modulus of elasticity (GPa)	Poisson's Ratio	CTE 10^(-6)* 1/(celsius)	Thermal Conductivity (W/mK)	Specific Heat	Electrical resistivity	Cost (USD/kg)	Relative Cost
Polycarbonate (PC)	62.1	2.2	1.20	2.38	0.36	122	0.20	840	2*10^14	Raw form: 0.8-5.30 Sheet: 2.50-4.00	Raw form: 3.4 Sheet: 4.2
Polystyrene (PS)	25-69	0.7-1.1	1.05	2.28-3.28	0.37-0.44	90-150	0.13	1170	>10^14	Raw form: 0.8-2.95	Raw form: 2.4
Poly(methyl methacrystalte) (PMMA)	53.8-73.1	0.7-1.6	1.19	2.24-3.24	0.33	90-162	0.17-0.25	1460	>10^12	Raw form: 0.8-3.60 Sheet: 2.00-3.80	Raw form: 2.7 Sheet: 3.8

1. Carbon fibers have high specific moduli and specific strengths.
2. They retain their high tensile modulus and high strength at elevated temperatures; high-temperature oxidation, however, may be a problem.
3. At room temperature, carbon fibers are not affected by moisture or a wide variety of solvents, acids, and bases.
4. These fibers exhibit a diversity of physical and mechanical characteristics, allowing composites incorporating these fibers to have specific engineered properties.
5. Fiber- and composite-manufacturing processes have been developed that are relatively inexpensive and cost effective.

Composites


Composites
Particle-reinforced
Fiber-reinforced
Structural
Nano
Large-particle
Dispersion-strengthened
Continuous (aligned)
Discontinuous (short)
Laminates
Sandwich panels
Aligned
Randomly oriented

- Injection molding
  - thermoplastic & some thermosets



Discontinuous and Aligned-Fiber Composites

Even though reinforcement efficiency is lower for discontinuous than for continuous fibers, discontinuous and aligned-fiber composites (Figure 16.8*b*) are becoming increasingly important in the commercial market. Chopped-glass fibers are used most extensively; however, carbon and aramid discontinuous fibers are also used. These short-fiber composites can be produced with moduli of elasticity and tensile strengths that approach 90% and 50%, respectively, of their continuous-fiber counterparts.

For a discontinuous and aligned-fiber composite having a uniform distribution of fibers and in which $l > l_c$, the longitudinal strength (σ^*_{cd}) is given by the relationship

For a discontinuous ($l > l_c$,) and aligned fiber–reinforced composite, longitudinal strength in tension

$$\sigma^*_{cd} = \sigma^*_f V_f \left(1 - \frac{l_c}{2l}\right) + \sigma'_m (1 - V_f) \qquad (16.18)$$

where σ^*_f and σ'_m represent, respectively, the fracture strength of the fiber and the stress in the matrix when the composite fails (Figure 16.9*a*).

If the fiber length is less than critical ($l < l_c$), then the longitudinal strength ($\sigma^*_{cd'}$) is given by

For a discontinuous ($l < l_c$) and aligned fiber–reinforced composite, longitudinal strength in tension

$$\sigma^*_{cd'} = \frac{l\tau_c}{d} V_f + \sigma'_m (1 - V_f) \qquad (16.19)$$

where d is the fiber diameter and τ_c is the smaller of either the fiber–matrix bond strength or the matrix shear yield strength.

Discontinuous and Randomly Oriented–Fiber Composites

For a discontinuous and randomly oriented fiber–reinforced composite, modulus of elasticity

Normally, when the fiber orientation is random, short and discontinuous fibers are used; reinforcement of this type is schematically demonstrated in Figure 16.8*c*. Under these circumstances, a *rule-of-mixtures* expression for the elastic modulus similar to Equation 16.10a may be used, as follows:

$$E_{cd} = KE_f V_f + E_m V_m \quad (16.20)$$

In this expression, K is a fiber efficiency parameter that depends on V_f and the E_f/E_m ratio. Its magnitude will be less than unity, usually in the range 0.1 to 0.6. Thus, for random-fiber reinforcement (as with oriented-fiber reinforcement), the modulus increases with increasing volume fraction of fiber. Table 16.2, which gives some of the mechanical

Table 16.2 Properties of Unreinforced and Reinforced Polycarbonates with Randomly Oriented Glass Fibers

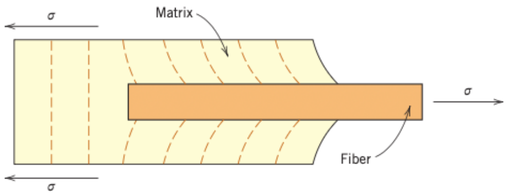
		Value for Given Amount of Reinforcement (vol%)		
Property	***Unreinforced***	***20***	***30***	***40***
Specific gravity	1.19–1.22	1.35	1.43	1.52
Tensile strength [MPa (ksi)]	59–62 (8.5–9.0)	110 (16)	131 (19)	159 (23)
Modulus of elasticity [GPa (10^6 psi)]	2.24–2.345 (0.325–0.340)	5.93 (0.86)	8.62 (1.25)	11.6 (1.68)
Elongation (%)	90–115	4–6	3–5	3–5
Impact strength, notched Izod (lb_f/in.)	12–16	2.0	2.0	2.5

Source: Adapted from Materials Engineering's *Materials Selector,* copyright © Penton/IPC.

Critical Fibre Length (lc)

$$l_c = \frac{\sigma_f^* d}{2\tau_c}$$

Ensure applied load is transmitted to the fibers by the matrix phase



shear yield strength of matrix: τ = 22.8 MPa[1]

Fiber diameter: $\mathrm{d} = 7\ \mu m$[2]

UTS: $\sigma_{f-rCF}^* = 3{,}300 MPa$ [2]

UTS: $\sigma_{f-CF}^* = 4{,}200\ MPa$ [3]

$$l_c = \frac{\sigma_{f-rCF}^* d}{2\tau c} = \frac{3300\ MPA * 7\mu m}{2.00 * 22.8\ MPa} = 507\ \mu m$$

$$l_c = \frac{\sigma_{f-CF}^* d}{2\tau c} = \frac{4200\ MPA * 7\mu m}{2.00 * 22.8\ MPa} = 645\ \mu m$$

[1] Cantrell et al. 2016 [2] Semitekolos et al. 2024 [3] Gao et al. 2023

3D-printed fibres are too short, critical length not reached

- Nozzle size is $400\ \mu m$, $L < l_c$
- Avg. fiber length $\mathrm{L} = 300\ \mu m$

$$\sigma_f^* = \frac{L \cdot 2\tau_c}{d} \qquad \sigma_f^* = \frac{300\mu m \cdot 2 \cdot 22.8MPa}{7\mu m} = 1{,}954\ MPa$$

Effective strength: ~1,954 MPa;

Potential Wasted: ~1,346 MPa wasted.

Ultimate Longitudinal Fracture Strength

Discontinuous, aligned-fiber, $L < l_c$ the formula is:

$$\sigma^*_{cd'} = \frac{L\tau_c}{d}V_f + \sigma_m{}'(1 - V_f)$$

Assuming $V_f = 0.07$ *of CF .* σ_m *: Callister lists Matrix/PC yield to be* **62.1 MPa**

$$\sigma^*_{cd'} = \frac{300\mu m \cdot 22.8MPa}{7\mu m} \cdot 0.07 + 62.1\ \mathrm{MPa}(1 - 0.07) = 126\ MPa$$

Prusa reported Tensile Strength 70 ± 3 MPa [1]
RTP reported Tensile Strength 107 MPa [2]

[1] Prusa Research (n.d.) [2] RTP Company (n.d.)

Estimating Elongation

For calculating elasticity of fibers which are discontinuous and randomly oriented:

$$E_{cd} = KE_f V_f + E_m V_m$$

K how randomly aligned are the fiber (low-high: 0-1), Assuming $V_f = 0.07$,

$$E_{cd} = K230 * 0.07 + 2.38 * 0.93 = 16.1K + 2.2134$$

According to Callister the K value typically ranges from 0.1 to 0.6.

$$E_{cd} = 16.1 * 0.1 + 2.2134 = 3.82\ GPa$$

$$E_{cd} = 16.1 * 0.6 + 2.2134 = 11.9\ GPa$$

Datasheet → Prusa: $E_{cd} = 3.2 \pm 0.1\ GPa$, RPT: $\mathrm{E_{cd}} = 7.58\ GPa$

Callister & Rethwisch (2018)

Estimating K-Factor

Finding the K-value using given elongation values:

Prusa: $3.2 = 16.1K + 2.2134 \rightarrow K = 0.061,$ RTP: $7.58 = 16.1K + 2.2134 \rightarrow K = 0.33$

According to Calister k values:

- ~0.2 fibers are randomly and uniformly distributed within three dimensions.
- ~0.375 fibers randomly and uniformly distributed within a specific plane